1. Introduction

- Image segmentation is an important stage in the road damage detection system. After enhancing video frames, segmentation is used to separate damaged road areas from the normal road surface. This helps to clearly identify potholes, cracks, and other road defects.
- Segmentation is required after enhancement because improved image quality (better contrast and reduced noise) allows more accurate detection of damaged regions. Without this step, it is difficult to isolate defects clearly due to unclear or noisy frames.
- The main goal of segmentation is to extract only the road damage areas for further analysis. This improves the accuracy of detecting and classifying different types of road deterioration.
- Good segmentation directly affects damage severity assessment because correctly identified regions allow better measurement of size and shape of defects. This helps in classifying damage as minor, moderate, or severe.
- Overall, segmentation connects the enhancement and evaluation stages and plays a key role in producing accurate final results in the road damage detection system.

2. Proposed Frame Segmentation Pipeline (Core Section)

❖ **Pipeline**

```
Video Frame
↓
Grayscale Conversion
↓
Gaussian Blur
↓
Edge Detection
↓
Morphological Processing
↓
Contour Detection
↓
Damage Classification
↓
Final Output
```

1. Input Enhanced Frame

Purpose:
To use pre-processed frames obtained from the enhancement stage.

Theory:
Enhanced frames contain improved contrast and reduced noise, making segmentation more effective.

Input: Enhanced Road frame
Output: Ready frame for processing

Contribution:
Provides a clean input for accurate segmentation.

2. Grayscale Conversion

Purpose:
To simplify image data by converting RGB images into grayscale.

Theory:
Reduces computational complexity by removing color information while preserving intensity values.

Formula:
$Gray = 0.299R + 0.587G + 0.114B$

Input: Colored frame
Output: Grayscale image

Contribution:
Prepares image for thresholding and edge detection.

3. Noise Reduction (Filtering)

Purpose:
To remove unwanted noise from the image.

Algorithm Used: Gaussian Blur

Theory:
Gaussian filtering smooths the image using weighted averaging of neighboring pixels.

Input: Grayscale image
Output: Smoothed image

Contribution:
Improves accuracy of segmentation by reducing false edges.

4. Thresholding / Edge Detection

Purpose:
To separate damaged regions from the background road surface.

Algorithms Used:

- Canny Edge Detection or
- Binary Thresholding

Theory:
Identifies sharp intensity changes that represent edges or damage boundaries.

Input: Smoothed image
Output: Binary edge map

Contribution:
Highlights potential damaged areas.

5. Morphological Processing

Purpose:
To refine detected regions and remove small noise.

Algorithms Used:

- Dilation
- Erosion
- Opening and Closing

Theory:
Morphological operations modify binary images based on shape structures.

Input: Binary edge image
Output: Cleaned segmented image

Contribution:
Improves region quality and fills gaps in detected damage.

6. Contour Detection

Purpose:
To identify boundaries of damaged regions.

Algorithm Used: OpenCV Contour Detection

Theory:
Contours are curves joining all continuous points along a boundary with same intensity.

Input: Processed binary image
Output: Detected contours

Contribution:
Helps locate exact damage areas.

7. Damage Region Extraction

Purpose:
To isolate and mark damaged road areas.

Output:
Final segmented image with highlighted damage regions (bounding boxes or shapes)

Contribution:
Provides final input for severity analysis.

https://github.com/MIFN08/Road-Damage-Detection.git

3. Experimental Setup and Results

3.1 Test Data Preparation

- Number of video frames: 42
- Frame size: 800 × 500
- Frame selection: Every 15th frame
- Damage types: Crack, pothole, alligator crack

3.2 Evaluation Metrics

- **<u>Quantitative Evaluation Metrics</u>**

Accuracy

Accuracy measures the overall correctness of the segmentation model in identifying damaged and non-damaged regions.

Accuracy=TP+TN/TP+TN+FP+FN

Precision

Precision measures how many detected damaged pixels are actually correct.

Precision=TP/TP+FP

Recall

Recall measures the ability of the model to detect actual damaged regions.

Recall=TP/TP+FN

F1-Score

F1-score provides a balance between precision and recall.

F1=2TP/2TP+FP+FN

Intersection over Union (IoU)

IoU measures the overlap between the predicted segmented region and the ground truth region.

IoU=Intersection/Union

Dice Coefficient

Dice coefficient measures segmentation similarity between predicted and ground truth masks.

Dice=2|A∩B|/|A|+|B|

```python
pred = cv2.imread("results/mask/mask_0.jpg", 0)

# Load ground truth mask
gt = cv2.imread("ground_truth/gt_0.png", 0)

# Resize if needed
gt = cv2.resize(gt, (pred.shape[1], pred.shape[0]))

# Binary conversion
_, pred = cv2.threshold(pred, 127, 1, cv2.THRESH_BINARY)
_, gt = cv2.threshold(gt, 127, 1, cv2.THRESH_BINARY)

# Calculate TP, TN, FP, FN
tp = np.sum((pred == 1) & (gt == 1))
tn = np.sum((pred == 0) & (gt == 0))
fp = np.sum((pred == 1) & (gt == 0))
fn = np.sum((pred == 0) & (gt == 1))

# Metrics
accuracy = (tp + tn) / (tp + tn + fp + fn)

precision = tp / (tp + fp + 1e-6)

recall = tp / (tp + fn + 1e-6)

f1 = (2 * precision * recall) / (precision + recall + 1e-6)

iou = tp / (tp + fp + fn + 1e-6)

dice = (2 * tp) / ((2 * tp) + fp + fn + 1e-6)

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-score :", round(f1, 4))
print("IoU      :", round(iou, 4))
print("Dice     :", round(dice, 4))
```

```
Accuracy : 0.9293
Precision: 0.0795
Recall   : 0.1833
F1-score : 0.1109
IoU      : 0.0587
Dice     : 0.1109
```

3.4 Qualitative Testing and Visual Results

Orginal_Frame

Enhanced_Frame

Mask_Frame

Detected_Frame

6. Contribution Table

Member Name (Reg No / Index No)	Segmentation Responsibility (Segmented Degradation Type)	Coding Contribution	Experimental Task
P.P. K. BHAGYA ICT/2023/101	Degradation 01 (Crack Type01)	Part of the Segmentation Pipeline Developed	Test on selected crack frames
S. PAVATHARAN ICT/2023/102	Degradation 02 (Crack Type2)	Part of the Segmentation Pipeline Developed	Test on selected crack frames
M.I.F. NUHA ICT/2023/103	Degradation 3 (Pothole)	Segmentation Pipeline Developed	Test on pothole frames
M.F.A. AFNAN ICT/2023/104	Overall dataset preparation	Developed full segmentation pipeline integrating	Executed full pipeline testing on all frames